

MATH 10A — STRUCTURE AND CONCEPTS IN MATH I

## §4.3

# GCF and LCM

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Greatest common factor and least common multiple

# Today's Goal

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After completing this section, you should be able to:

- State what the greatest common factor and the least common multiple of a set of whole numbers are, and read and write the notation  $\text{GCF}(a, b)$  and  $\text{LCM}(a, b)$
- Find the GCF of a set of whole numbers by the brute force, prime factorization, and repeated division methods
- Find the LCM of a set of whole numbers by the brute force, prime factorization, and repeated division methods

# Today's Goal, Continued

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- Use the relationship  $a \cdot b = \text{GCF}(a, b) \cdot \text{LCM}(a, b)$  to find either one from the other
- Apply the Euclidean Algorithm to find the GCF of two large numbers, and explain why the algorithm works

# Learning Target 12

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## LT.12 GCF and LCM

I can apply multiple strategies to find the greatest common factor and least common multiple of a set of whole numbers.

# Factors of 72 and 48

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List all the factors of 72. List all the factors of 48. Which factors do they have in common?

- **Factors of 72:** 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72
- **Factors of 48:** 1, 2, 3, 4, 6, 8, 12, 16, 24, 48

The factors they have in common are 1, 2, 3, 4, 6, 8, 12, and 24.

Which of these is the largest?

# The Greatest Common Factor

## Definition 4.3.1

The **greatest common factor** (GCF) of two or more whole numbers is the largest factor that those numbers have in common.

For two numbers,  $a$  and  $b$ , we denote the GCF of  $a$  and  $b$  as  $\text{GCF}(a, b)$ .

$$\text{GCF}(72, 48) = 24$$

# Brute Force Method for GCF

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## Procedure 4.3.2 Brute Force Method for GCF

1. List all factors of each number for which we want to find the GCF.
2. Identify the factors that these numbers have in common.
3. The GCF will be the largest number from Step 2.

This is the method we just used for 72 and 48.

# Prime Factorization Method for GCF

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## Procedure 4.3.4 Prime Factorization Method for GCF

1. Find the prime factorization of each number for which we want to find the GCF.
2. Identify the prime factors that these numbers have in common.
3. The GCF will be the product of the prime factors from Step 2.

# GCF by Prime Factorization

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Use the prime factorization method to find  $\text{GCF}(72, 48)$ .

The prime factorizations are

$$72 = 2^3 \cdot 3^2$$

$$48 = 2^4 \cdot 3$$

We see that 72 and 48 have three factors of 2 and one factor of 3 in common. So

$$\text{GCF}(72, 48) = 2^3 \cdot 3 = 24.$$

# Repeated Division Method for GCF

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## Procedure 4.3.6 Repeated Division Method for GCF

1. Choose a factor that the numbers have in common and divide each number by this factor.
2. Repeat Step 1 until the quotients have no common factors.
3. The GCF will be the product of all the factors you divided by in Step 1.

## GCF by Repeated Division

Use the repeated division method to find  $\text{GCF}(72, 48)$ .

|   |    |    |
|---|----|----|
| 2 | 72 | 48 |
| 2 | 36 | 24 |
| 6 | 18 | 12 |
|   | 3  | 2  |

Since 3 and 2 do not have any common factors, we stop. This gives us

$$\text{GCF}(72, 48) = 2 \cdot 2 \cdot 6 = 24.$$

# The Least Common Multiple

## Definition 4.3.8

The **least common multiple** (LCM) of two or more whole numbers is the smallest multiple that those numbers have in common.

For two numbers,  $a$  and  $b$ , we denote the LCM of  $a$  and  $b$  as  $\text{LCM}(a, b)$ .

# Brute Force Method for LCM

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## Procedure 4.3.9 Brute Force Method for LCM

1. List multiples of each number for which we want to find the LCM. Note that you cannot create a complete list since each list of multiples will continue indefinitely.
2. Identify the multiples that these numbers have in common.
3. The LCM will be the smallest number from Step 2.

# LCM by Brute Force

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Use the brute force method to find  $\text{LCM}(72, 48)$ .

- **Multiples of 72:** 72, 144, 216, 288, ...
- **Multiples of 48:** 48, 96, 144, 192, 240, 288, ...

The multiples they have in common are 144, 288, ... Therefore,  
 $\text{LCM}(72, 48) = 144$ .

# Prime Factorization Method for LCM

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## Procedure 4.3.11 Prime Factorization Method for LCM

1. Find the prime factorization of each number for which we want to find the LCM.
2. Identify each distinct prime factor from Step 1.
3. The LCM will be the product of the prime factors from Step 2, each raised to the greatest exponent that occurs in the prime factorizations from Step 1.

# LCM by Prime Factorization

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Use the prime factorization method to find  $\text{LCM}(72, 48)$ .

$$72 = 2^3 \cdot 3^2$$

$$48 = 2^4 \cdot 3$$

The distinct prime factors that are needed are 2 and 3.

Notice that 72 requires three factors of 2 while 48 requires four factors of 2. This means that the LCM will need four factors of 2.

Similarly, the LCM will need two factors of 3. So  $\text{LCM}(72, 48) = 2^4 \cdot 3^2 = 144$ .

# Repeated Division Method for LCM

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## Procedure 4.3.13 Repeated Division Method for LCM

1. Choose a prime factor that at least two of the numbers have in common and divide each number by this factor (if they are divisible by this factor).
2. Repeat Step 1 until none of the quotients have any common factors.
3. The LCM will be the product of all the factors you divided by in Step 1 with the quotients that you ended up with that have no common factors.

# LCM by Repeated Division

Use the repeated division method to find  $\text{LCM}(72, 48)$ .

|   |    |    |
|---|----|----|
| 2 | 72 | 48 |
| 2 | 36 | 24 |
| 2 | 18 | 12 |
| 3 | 9  | 6  |
|   | 3  | 2  |

Since 3 and 2 have no common factors, we stop. Therefore,

$$\text{LCM}(72, 48) = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 2 = 144.$$

# GCF and LCM of 90 and 60

Find  $\text{GCF}(90, 60)$  and  $\text{LCM}(90, 60)$  by all three methods. The table shows which members of your group do each method.

| Method              | Group of 2 | Group of 3 | Group of 4 |
|---------------------|------------|------------|------------|
| Brute force         | A and B    | 1          | 1 and 2    |
| Prime factorization | A          | 2          | 3          |
| Repeated division   | B          | 3          | 4          |

Explain your method to your group. Did every method give the same answers?

# GCF and LCM of 90 and 60, Continued

$$\text{GCF}(90, 60) = 30$$

$$\text{LCM}(90, 60) = 180$$

|   |    |    |
|---|----|----|
| 2 | 90 | 60 |
| 3 | 45 | 30 |
| 5 | 15 | 10 |
|   | 3  | 2  |

Look at the repeated division method for 90 and 60. Where do you see the GCF? Where do you see the LCM?

# Two Numbers with GCF 6 and LCM 72

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**(a)** Find two whole numbers whose GCF is 6 and whose LCM is 72. Can your group find more than one pair?

6 and 72 work. So do 18 and 24.

**(b)** Multiply the two numbers in each pair. Then multiply the GCF and the LCM. What do you notice?

$6 \cdot 72 = 432$ ,  $18 \cdot 24 = 432$ , and  $\text{GCF} \cdot \text{LCM} = 6 \cdot 72 = 432$ .

# Product of GCF and LCM

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## Theorem 4.3.15 Product of GCF and LCM

Given any two whole numbers,  $a$  and  $b$ ,

$$a \cdot b = \text{GCF}(a, b) \cdot \text{LCM}(a, b).$$

For 72 and 48:  $72 \cdot 48 = 3456$  and  $24 \cdot 144 = 3456$ .

## GCF and LCM of 126 and 70

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Find  $\text{GCF}(126, 70)$  and  $\text{LCM}(126, 70)$ .

The prime factorizations are

- $126 = 2 \cdot 3^2 \cdot 7$
- $70 = 2 \cdot 5 \cdot 7$

Therefore,

$$\text{GCF}(126, 70) = 2 \cdot 7 = 14.$$

Using Theorem 4.3.15, we have

$$126 \cdot 70 = \text{GCF}(126, 70) \cdot \text{LCM}(126, 70)$$

$$8820 = 14 \cdot \text{LCM}(126, 70)$$

$$\frac{8820}{14} = \text{LCM}(126, 70)$$

$$630 = \text{LCM}(126, 70)$$

# The Euclidean Algorithm

## Procedure 4.3.17 The Euclidean Algorithm

1. Begin with two nonzero whole numbers  $a$  and  $b$ . If  $a = b$ , then  $\text{GCF}(a, b) = a$ . Otherwise, continue to the next step.
2. Divide the larger number by the smaller number. If the remainder is zero, then stop and go to Step 4. If the remainder is not zero, continue to Step 3.

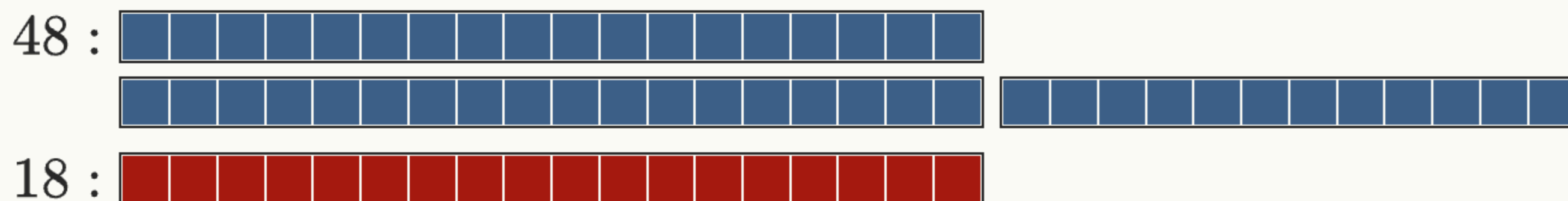
# The Euclidean Algorithm, Continued

## Procedure 4.3.17 The Euclidean Algorithm, continued

3. Divide the last divisor by the remainder. If the new remainder is zero, then stop and go to Step 4. If the new remainder is not zero, repeat this step.
4. The GCF will be the last divisor that resulted in a remainder of zero.

# Why the Euclidean Algorithm Works

Suppose that we want to find  $\text{GCF}(48, 18)$ . Let  $d = \text{GCF}(48, 18)$ .



Since  $d$  is a divisor of both 48 and 18, it follows that  $d$  must also be a divisor of  $48 - 18 = 30$  and  $30 - 18 = 12$ .

In other words, we removed all groups of 18 from 48. The remainder, 12, must still be divisible by  $d$ .

# Why It Works, Continued

Now we need to find  $\text{GCF}(18, 12)$ .

18 : 

12 : 

Since  $d$  is a divisor of both 18 and 12, it follows that  $d$  must also be a divisor of  $18 - 12 = 6$ .

In other words, we removed all groups of 12 from 18. The remainder, 6, must still be divisible by  $d$ .

# Why It Works, Continued

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Now we need to find  $\text{GCF}(12, 6)$ .

12 : 

6 : 

Since  $d$  is a divisor of both 12 and 6, it follows that  $d$  must also be a divisor of  $12 - 6 = 6$  and  $6 - 6 = 0$ .

In other words, we removed all groups of 6 from 12. Now we need to find  $\text{GCF}(6, 0)$ , which is just 6.

# Find the GCF of 110 and 40

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Use the Euclidean Algorithm to find  $\text{GCF}(110, 40)$ .

$$110 \div 40 \quad 110 = 2 \cdot 40 + 30 \quad \text{So } \text{GCF}(110, 40) = \text{GCF}(40, 30)$$

$$40 \div 30 \quad 40 = 1 \cdot 30 + 10 \quad \text{So } \text{GCF}(40, 30) = \text{GCF}(30, 10)$$

$$30 \div 10 \quad 30 = 3 \cdot 10 + 0 \quad \text{So } \text{GCF}(30, 10) = \text{GCF}(10, 0)$$

Since the remainder is 0, we can stop and we get

$$\text{GCF}(110, 40) = \text{GCF}(40, 30) = \text{GCF}(30, 10) = \text{GCF}(10, 0) = 10.$$